

MycoFlow — Biologically Inspired, Reflexive QoS for OpenWrt

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GRADUATION PROJECT · 2025-2026

MOTIVATION

Problem

- Bufferbloat:** A single TCP flow causes 693 ms queue delay on a 20 Mbit/s WAN connection — making gaming and VoIP unusable.
- Static QoS:** Even modern AQMs like CAKE rely on DSCP tags; ISPs strip these tags, causing all traffic to fall into the same queue.
- Fixed Bandwidth:** WAN speed limit is configured once during setup; it cannot react to real-time congestion signals.

APPROACH

Solution

- 1 Biological Inspiration:** Adaptive nutrient distribution strategy of fungal mycelium networks → reflexive network control
- 2 6-Class Device Persona:** VOIP, GAMING, VIDEO, STREAMING, BULK, TORRENT — On-chip <1 μs decision without DPI or ML
- 3 12-Class Flow Taxonomy:** 3-signal weighted voting combining passive DNS (0.6), target port hints (0.3), and behavioral fingerprinting (0.1)
- 4 Zero Flash Writes:** All state is kept in RAM — no damage to NAND flash lifespan (live tested on Xiaomi AX3000T)

COMPARISON

System	Embedded	Automatic	No ML	No DPI	Adaptive
SQM/CAKE	✓	✗	✓	✓	✗
qosify	✓	✗	✓	✗	✗
nDPI	✗	✓	✗	✗	✗
DeepFlow	✗	✓	✗	✗	✗
MycoFlow	✓	✓	✓	✓	✓

DSCP → CAKE TIN MAPPING

VOIP	→ EF → Voice tin	GAMING	→ CS4 → Voice tin
VIDEO	→ CS3 → Video tin	STREAMING	→ CS2 → Video tin
BULK	→ CS1 → Bulk tin	TORRENT	→ CS1 → Bulk tin

SYSTEM SPECIFICATIONS

TARGET HARDWARE Xiaomi AX3000T (MT7981B, 256MB)	
CPU USAGE 6% median	RAM (RSS) 1,152 KB
TEST COVERAGE 13/13 CTest	DEPLOYMENT OpenWrt IPK

SYSTEM ARCHITECTURE

1 Hz Reflexive Control Loop



EXPERIMENTAL RESULTS

Core Metrics

154× Gaming tin latency advantage — Voice 225 μs vs Bulk 34.7 ms	38% Gaming latency vs. CAKE-alone — 5-persona load (1.43 → 1.10 ms)
70.4% In-vivo tin-matched accuracy with passive DNS from 30% — N=1,650 flows	1,152 KB Constant RSS · zero NAND flash writes · 6% median CPU (A53)

PHASE 1 — DUAL PERSONA VALIDATION (20 MBIT/S WAN)		
FIFO (no QoS)		693 ms
CAKE (standalone)		0.63 ms
CAKE + MycoFlow		0.71 ms (+0.08ms overhead)
PHASE 2 — SIX PERSONA MIXED LOAD (5 CLIENTS)		
CAKE (5 clients)		1.43 ms
MycoFlow (5 clients)		1.10 ms -38% gaming latency
IN-VIVO DNS VALIDATION — TIN-MATCHED ACCURACY (N=1,650 Flows · seed=42)		
No DNS (port+behav.)		49.26%
DNS active (80-cat)		66.97%
DNS active (585-cat)		70.41%

Future Work: Sub-second eBPF sensing · cooperative mycelium maps · TinyML classification.

Conclusion: MycoFlow proves bio-inspired reflexive QoS on consumer CPE: 154× gaming advantage, 38% latency reduction under mixed load, 70.4% CAKE tin-matched accuracy with passive DNS — all at 1,152 KB RAM, 6% CPU, zero flash writes.

